Digoxin Intravenous

FORM Ampoule containing 500 micrograms/2 ml solution

INDICATION Control of specific arrhythmias as per cardiology

LOADING DOSE: Administer in divided doses over 12 hours as detailed below. Assess the clinical response before giving each additional dose.

AGE	DOSE	FREQUENCY	ROUTE
Neonate <1.5 kg	10 microgram/kg	Initial dose	IV infusion
	5 microgram/kg	Second dose after 6 hours	IV infusion
	5 microgram/kg	Third dose after a further 6 hours	IV infusion
Neonate 1.5 kg to 2.5 kg	15 microgram/kg	Initial dose	IV infusion
	7.5 microgram/kg	Second dose after 6 hours	IV infusion
	7.5 microgram/kg	Third dose after a further 6 hours	IV infusion
Neonate > 2.5 kg or infant 1 to 6 months	17.5 microgram/kg	Initial dose	IV infusion
	8.75 microgram/kg	Second dose after 6 hours	IV infusion
	8.75 microgram/kg	Third dose after a further 6 hours	IV infusion

MAINTENANCE DOSE: Administer 12 hours after third loading dose

NB: Careful clinical observation and monitoring of serum digoxin levels should be used as a basis for adjustment of dosage.

AGE	DOSE	FREQUENCY	ROUTE
Neonate <1.5 kg	2 to 3 microgram/kg	12 hourly	IV infusion
Neonate 1.5 kg to 2.5 kg	2 to 3 microgram/kg	12 hourly	IV infusion
Neonate > 2.5 kg or infant 1 to 6 months	5 microgram/kg	12 hourly	IV infusion
> 6 months	See paediatric monographs		

*Note: The total daily dose may be administered once daily. Adjust dose accordingly in renal impairment.

RECONSTITUTION Already in solution

DILUTION May be given undiluted if appropriate.

Dilute as follows for small doses

Digoxin 500 micrograms/2 ml	1 ml
Compatible infusion solution (see below) e.g Glucose 5%	Up to 25 ml total

Gives a 10 micrograms/ml solution (250mcg/25ml). Use required volume.

Dilution should be carried out immediately before use and any unused solution discarded.

METHOD OF ADMINISTRATION	Loading dose: IV infusion via pump over 30-60 minutes Maintenance dose: IV infusion via pump over 10-20 minutes Avoid rapid intravenous administration (risk of hypertension and reduced coronary flow)

COMPATIBILITY

Solution compatibility	Glucose 5%, Sodium chloride 0.45%, Sodium chloride 0.9%
Solution incompatibility	
IV Line compatibility	Furosemide (in sodium chloride only), Heparin Sodium, Hydrocortisone, Linezolid, Midazolam, Milrinone, Morphine Sulphate, Potassium Chloride
IV Line incompatibility	Amiodarone, Fluconazole, Insulin

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
- See Summary of Product Characteristics and most recent edition of BNF for Children

FURTHER INFORMATION

- A plasma digoxin level should be taken 6 hours after a dose and concentration maintained in the range of 0.8 - 2micrograms/L is usually recommended (see product literature for further information). Approximate time to steady state is 5-10 days but levels may be indicated earlier to observe for signs of toxicity, especially with concomitant prescriptions.
- Monitor electrolytes and renal function
- The clinical effect of digoxin is potentiated by amiodarone. The maintenance dose of digoxin should be halved if any of these drugs are also prescribed.
- The intramuscular (IM) route is painful and associated with tissue necrosis and therefore cannot be recommended.
- Bioavailability differs between IV and oral digoxin so ensure plasma monitoring when changing formulation.
- All digoxin products contain ethanol and propylene glycol. There is a risk of adverse effects. Monitor infant appropriately
- Digoxin has a high osmolarity and may cause venous irritation and tissue damage in cases of extravasation. If a central venous access device is unavailable, administer via a large peripheral vein monitoring insertion site closely. Re-site cannula at first signs of inflammation

PH 6.7 - 7.3 (undiluted)

LICENSED STATUS Licensed for use in all ages.

APPLICABLE POLICIES West of Scotland Neonatal Guidelines

Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department